Building intuition about exterior powers

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Stop. Pause. Breathe.

Ask yourself:

> Why are we dealing with alternating multilinear maps in the first place? Why are they so special?

In the previous notes we have defined alternating multilinear maps as those maps which respect a certain action of S_n (i.e., they respect a certain symmetry in the domain).

From there we defined $\Lambda^n(V)$ as the largest quotient space on which any linear map corresponds to an alternating multilinear map on $\underbrace{(V, ..., V)}_{n}$.

Now, let us look at it in the **reverse direction**. We know that alternating maps can't distinguish between two inputs if they project onto the same wedge product in $\Lambda^n(V)$. So, hopefully, figuring out when two inputs have the same wedge product should give us some idea about alternating maps.

Q. For some $1 \leq k \leq n$, given two k sized sets of linearly independent vectors in $\mathbb{R}^n$, when do the have the same wedge product? That is, given independent sets $\{x_1, ..., x_k\} \subseteq \mathbb{R}^n$ and $\{y_1, ..., y_k\} \subseteq \mathbb{R}^n$, when do we have

$$x_1 \wedge ... \wedge x_k = y_1 \wedge ... \wedge y_k?$$

A. Since $x_1, ..., x_k$ are all linearly independent, so we can find vectors $x_{k+1}, ..., x_n$ such that $(x_1, ..., x_n)$ forms a basis for $\mathbb{R}^n$.

So we have a full-rank matrix $A_{n \times k}$ such that

$$[x_1 \ \ ... \ \ x_n]A = [y_1 \ \ ... \ \ y_k]$$

Recall that $\left\{x_{i_1} \wedge ... \wedge x_{i_k} : 1 \leq i_1 < ... < i_k \leq n\right\}$ is a basis for $\Lambda^k(\mathbb{R}^n)$. The coordinate of $y_1 \wedge ... \wedge y_k$ along $x_{i_1} \wedge ... \wedge x_{i_k}$ is the $k \times k$ minor of A on the rows $i_1, ..., i_k$ and across all the columns.

Since $y_1 \wedge ... \wedge y_k$ only has coordinate along the basis vector $x_1 \wedge ... \wedge x_k$, so only the top $k \times k$ minor of A is non-zero, all others $k \times k$ minors are zero. Therefore, the top k-many rows of A are linearly independent. Every other row must be 0, or else we can swap one of the top k-many rows with this row to produce another k-sized set of rows which are linearly independent, which is a contradiction.

Since all the rows of A except the top k rows are zero, so define A' as the top $k \times k$ submatrix of A. We have that

$$[x_1 \ \ ... \ \ x_k]A' = [y_1 \ \ ... \ \ y_k],$$

and $\det(A') = 1$.

Therefore, $y_1, ..., y_k$ are in the span of $x_1, ..., x_k$. Since $y_1, ..., y_k$ are linearly independent, so we get that

$$\text{span}(\{x_1, ..., x_k\}) = \text{span}(\{y_1, ..., y_k\}).$$

In fact, this is a characterization of the wedge products being same.

So, we get that given linearly independent sets $\{x_1, ..., x_k\}, \{y_1, ..., y_k\} \subseteq \mathbb{R}^n$,

$$x_1 \wedge ... \wedge x_k = y_1 \wedge ... \wedge y_k \Leftrightarrow$$
$$\exists A_{k\times k} \text{ s.t.} \det(A) = 1 \text{ and } [x_1 \ ... \ x_k]A = [y_1 \ ... \ y_k].$$

Observe that the existence of such a matrix A implies that $\text{span}(\{x_1, ..., x_k\}) = \text{span}(\{y_1, ..., y_k\})$.

Assume $1 \leq k \leq n$. We get some *very* interesting things from the above characterization:

- Every non-zero simple k-vector $v_1 \wedge ... \wedge v_k \in \Lambda^k(\mathbb{R}^n)$ describes the equivalence class of all k-dimensional parallelepipeds living in $\text{span}(\{v_1, ..., v_k\})$ and having the same signed k-volume as the oriented parallelepiped spanned by $(v_1, ..., v_k)$.
- The k-fold input to an alternating multilinear map can be thought of as a k-dimensional parallelepiped in $\mathbb{R}^n$. Therefore, elements of $\text{Hom}_{\mathbb{R}}\left(\Lambda^k(\mathbb{R}^n), V\right)$ are linear maps which cannot distinguish between two k-dimensional parallelepipeds which share the same span and have the same oriented volume.
- A map $L \in \text{Hom}_{\mathbb{R}}(\Lambda^2(\mathbb{R}^3), V)$ is uniquely determined by what it does to the parallelograms $e_1 \wedge e_2, e_2 \wedge e_3, e_1 \wedge e_3$, since every 2-dimensional parallelogram in $\mathbb{R}^3$ can be expressed as a linear combination of these 3 parallelograms.

Let us try to intuitively analyze $\varphi \in \text{Hom}_{\mathbb{R}}\left(\Lambda^k(\mathbb{R}^n), V\right)$, where V is some arbitrary vector space.

φ is uniquely determined by what it does to the "principal" k-parallelepipeds, namely $\left\{e_{i_1} \wedge ... \wedge e_{i_k} : 1 \leq i_1 < ... < i_k \leq n\right\}$. Observe that all of the principal k-parallelepipeds have absolute k-volume $= 1$.

Instead of dealing with abstract k, n, let us consider $\varphi \in \text{Hom}_{\mathbb{R}}(\Lambda^2(\mathbb{R}^3), V)$, the case of parallelograms in $\mathbb{R}^3$. φ is uniquely determined by the vectors (in V) it associates to the principal k-parallelepipeds $e_1 \wedge e_2, e_2 \wedge e_3, e_1 \wedge e_3$.

Given vectors $u = (u_1, u_2, u_3), v = (v_1, v_2, v_3) \in \mathbb{R}^3$, the parallelogram $\langle u, v\rangle$ casts shadows on the XY-,YZ- and XZ-planes. Now,

- We look at the first shadow, $\langle (u_1, u_2), (v_1, v_2)\rangle$ and check if it has the same orientation as $\langle e_1, e_2\rangle$.
- We calculate the (unsigned) area of $\langle (u_1, u_2), (v_1, v_2)\rangle$, and multiply it by $\varphi(e_1 \wedge e_2)$. If $\langle (u_1, u_2), (v_1, v_2)\rangle$ and $\langle e_1, e_2\rangle$ have different orientation, then we multiply by a further -1, otherwise we leave the product alone.
- We do the same for the other shadows and add all of the results together. This is the value of $\varphi(u \wedge v)$.

Observe that the above calculation matches with

$$\varphi(u \wedge v) = \varphi((u_1e_1 + u_2e_2 + u_3e_3) \wedge (v_1e_1 + v_2e_2 + v_3e_3)) = ...$$

Therefore $\varphi \in \text{Hom}_{\mathbb{R}}\left(\Lambda^k(\mathbb{R}^n), V\right)$ is uniquely determined by the vectors (in V) it assigns to unit k-volume parallelepipeds in every principal k-plane.

We are mostly interested in the case when $V = \mathbb{R}$, i.e., $\varphi \in \mathrm{Hom}_{\mathbb{R}}\big(\Lambda^k(\mathbb{R}^n), \mathbb{R}\big) \cong \Lambda^k(\mathbb{R}^n)^*$. From the above discussion we get that:

- Every $\varphi \in \mathrm{Hom}_{\mathbb{R}}\big(\Lambda^k(\mathbb{R}^n), \mathbb{R}\big)$ is uniquely determined by the **weight it assigns to projection onto every principal k-plane**, the plane spanned by k-many basis vectors.
- For example, for some $\varphi \in \mathrm{Hom}_{\mathbb{R}}(\Lambda^2(\mathbb{R}^3), \mathbb{R})$, φ associates the `XY`-, `YZ`- and the `XZ`-shadows with **weights**, and everything else about φ is uniquely determined from there.

Therefore, every element $\varphi \in \Lambda^k(\mathbb{R}^n)^*$ carries exactly the data of what weight it assigns to shadows onto each of the principal k-dimensional planes.

Playing around with the ideas developed above, we get that:

Every $\varphi \in \Lambda^k(\mathbb{R}^n)^*$ is a "coherent" way of assigning scalar values to k-dimensional parallelepipeds in $\mathbb{R}^n$ ("*measuring* a k-parallelepiped"), where "coherent" implies:

- φ should respect taking sum (as k-vectors) of k-parallelepipeds, whenever the sum is again a simple k-vector.
- For two k-parallelepipeds which have the **same span** in $\mathbb{R}^n$, if

$$\text{absolute-}k\text{-Volume}(A) = r \cdot \text{absolute-}k\text{-Volume}(B)$$

for some $r \in \mathbb{R}_{\geq 0}$, then

$$\varphi(A) = \begin{cases} r\varphi(B) \text{ if } A \text{ and } B \text{ have same orientation} \\ -r\varphi(B) \text{ otherwise} \end{cases}$$

Note that k-vectors refer to elements of $\Lambda^k(\mathbb{R}^n)$, and simple k-vectors are those which can be written as $v_1 \wedge \ldots \wedge v_k$ for some $v_i \in \mathbb{R}^n$.

This is the **intuitive view** of $\Lambda^k(\mathbb{R}^n)^*$ which I had been looking for. This will provide us with the much needed bridge to get to differential forms.

Using all that we have seen above, every $\varphi \in \Lambda^n(\mathbb{R}^n)^*$ is uniquely determined by where it sends $e_1 \wedge \ldots \wedge e_n$, i.e., φ is uniquely determined by what scalar it assigns to unit volume n-parallelepipeds in $\mathbb{R}^n$.

When we choose to assign $1 \in \mathbb{R}$ to unit volume n-parallelepipeds in $\mathbb{R}^n$, we get $\det_n \in \Lambda^n(\mathbb{R}^n)^*$.